

CSC 2400-002: Design of Algorithms (Spring 2026)

Tennessee Tech University

Drill 15 Solutions

Topic: Euclid's GCD Algorithm, Divide & Conquer, MergeSort, QuickSort

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Question 1. Explain why the pseudocode of Euclid's GCD algorithm seen in class (and given in the [lecture notes](#)) remains correct even if we remove lines 1-2.

Solution: For $a > b$, we have $r = b \bmod a = b$. Thus, even if we remove lines 1-2, $\text{GCD}(a, b)$ with $a > b$ recurses on $\text{GCD}(b, a)$, which effectively swaps a and b and we are ensured that the first number is smaller than the second number in the next recursive call.

Question 2. Run the recursive procedure of Euclid's GCD algorithm (not the explicit division method, but as shown in page 2 of the [lecture notes](#) under "Example runs") to compute the following. Show and justify the steps as in the notes.

1. $\text{GCD}(399, 54)$

2. $\text{GCD}(30, 462)$

Solution:

$$\begin{aligned}\text{GCD}(399, 54) &= \text{GCD}(21, 54) && \text{since } 399 \bmod 54 = 21 \\ &= \text{GCD}(12, 21) && \text{since } 54 \bmod 21 = 12 \\ &= \text{GCD}(9, 12) && \text{since } 21 \bmod 12 = 9 \\ &= \text{GCD}(3, 9) && \text{since } 12 \bmod 9 = 3 \\ &= \text{GCD}(0, 3) && \text{since } 9 \bmod 3 = 0 \\ &= 3\end{aligned}$$

$$\begin{aligned}\text{GCD}(30, 462) &= \text{GCD}(12, 30) && \text{since } 462 \bmod 30 = 12 \\ &= \text{GCD}(6, 12) && \text{since } 30 \bmod 12 = 6 \\ &= \text{GCD}(0, 6) && \text{since } 12 \bmod 6 = 0 \\ &= 6\end{aligned}$$

Question 3. Show how MergeSort will work on the following array using the binary-tree-like structure in the examples given in slide 8 in the [lecture slides](#) or in Figure 1 of the [lecture notes](#). Do mark the step numbers as in the examples.

[34, 87, 56, 22, 10, 99, 3, 1]

Solution:

